CSE 366 Artificial Intelligence

Submitted To

Dr. Raihan Ul Islam

Associate Professor

Submitted By

Munem Islam Prionto
2023-2-60-024

Mashrur Bin Morshed
2023-2-60-110

Fahmida Jannat
2025-1-60-334

Syeda Lamia
2022-3-60-315

What is PM2.5?

PM2.5 means Particulate Matter with diameter ≤ 2.5 micrometers.

PM value is the amount of very tiny air particles (≤ 2.5 micrometers) in the air

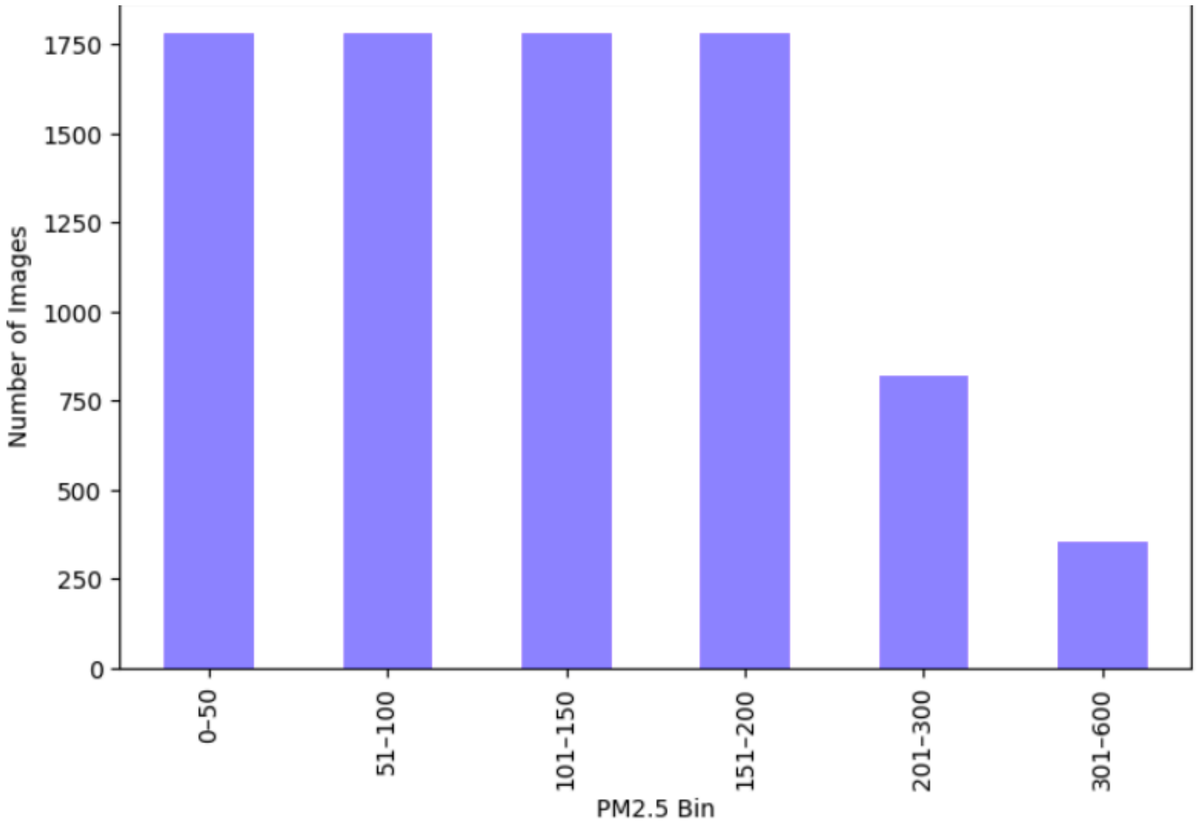
Dataset Overview

PM25Vision

A large-scale dataset for estimating air quality (PM2.5) from street-level images

Initial Observation

- First four classes are balanced
- Last two classes are imbalanced
- Total 11,000+ Images



Sample images (Class Wise)

0–50 (25.0)

51–100 (67.0)

101–150 (124.0)

151–200 (157.0)

201–300 (224.0)

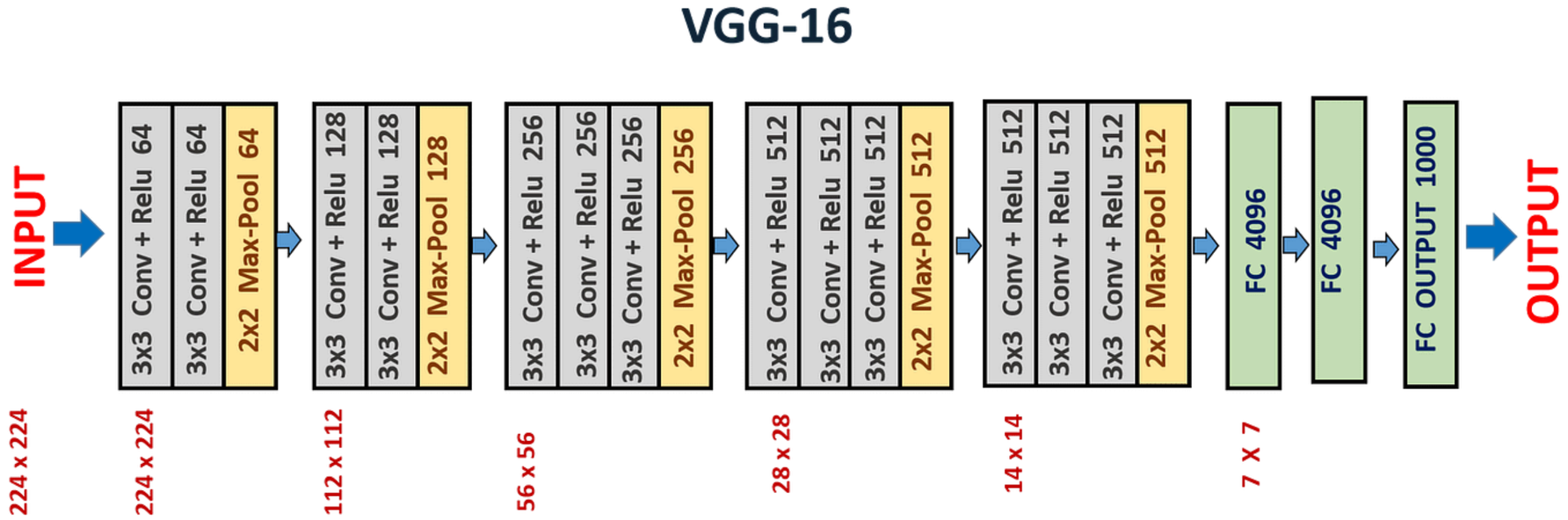
301–600 (445.0)

Models Applied

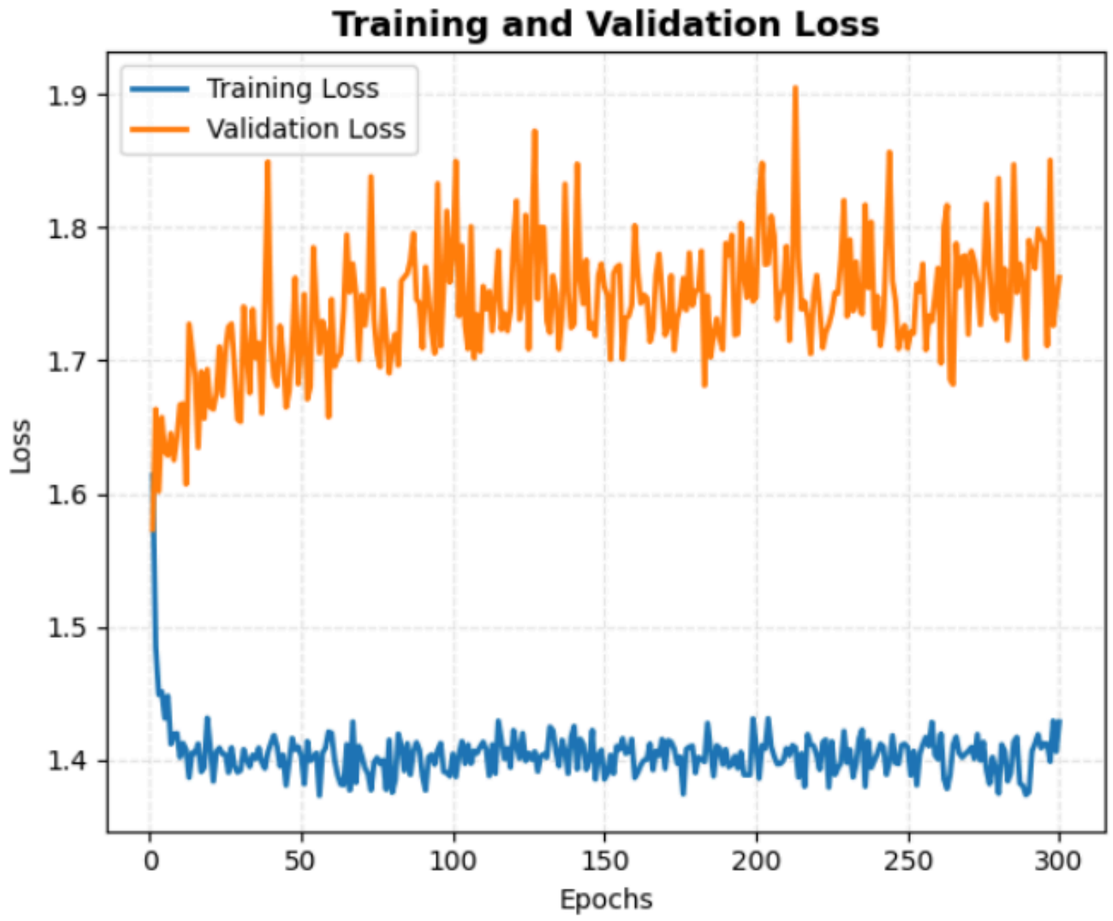
VGG-16

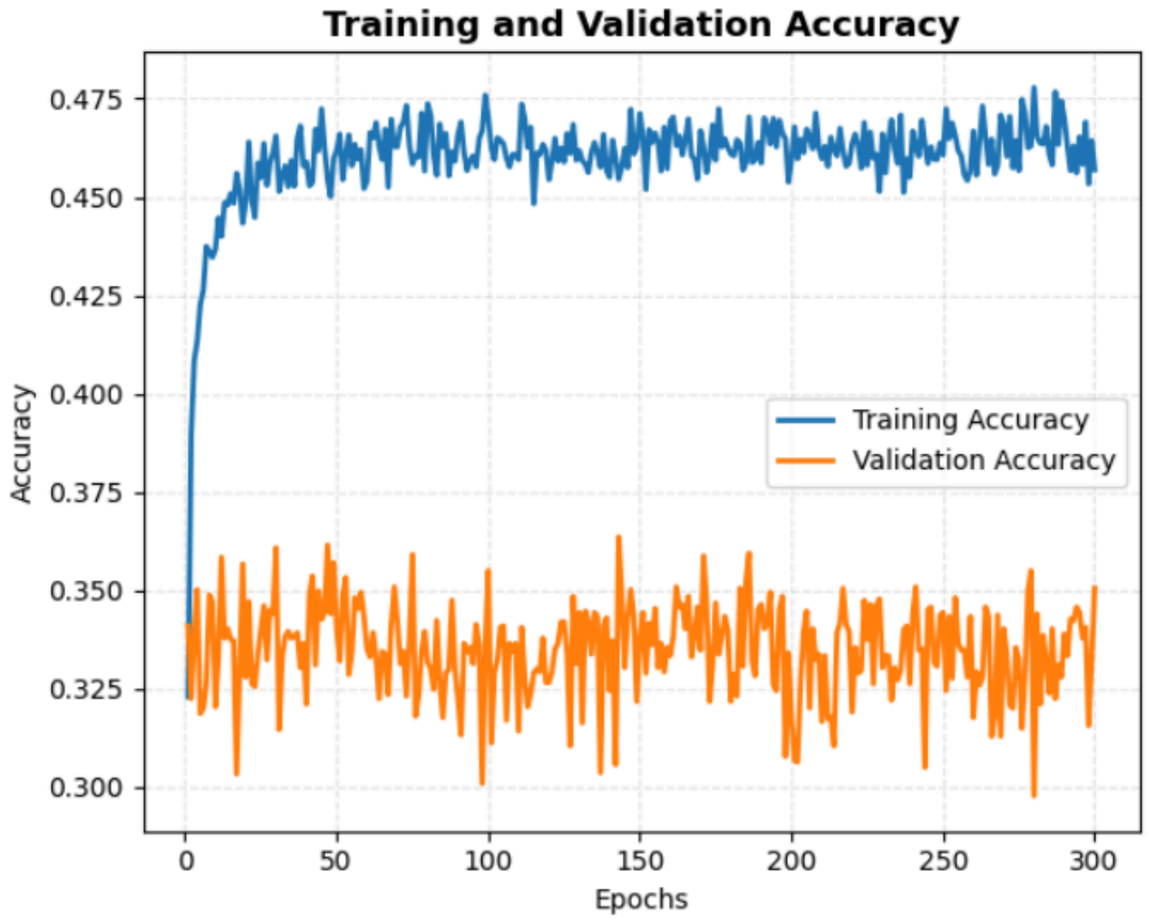
- 16 layers deep
- Simple stacked convolutional layers with uniform structure

Accuracy → 0.3506
Precision → 0.3100
Recall → 0.2889
F1-Score → 0.2777
AUC → 0.6831



VGG-16


Training and Validation Loss
Training Loss
Validation Loss
Loss
1.9
1.8
1.7
1.6
1.5
1.4
0
50
100
150
200
250
300
Epochs

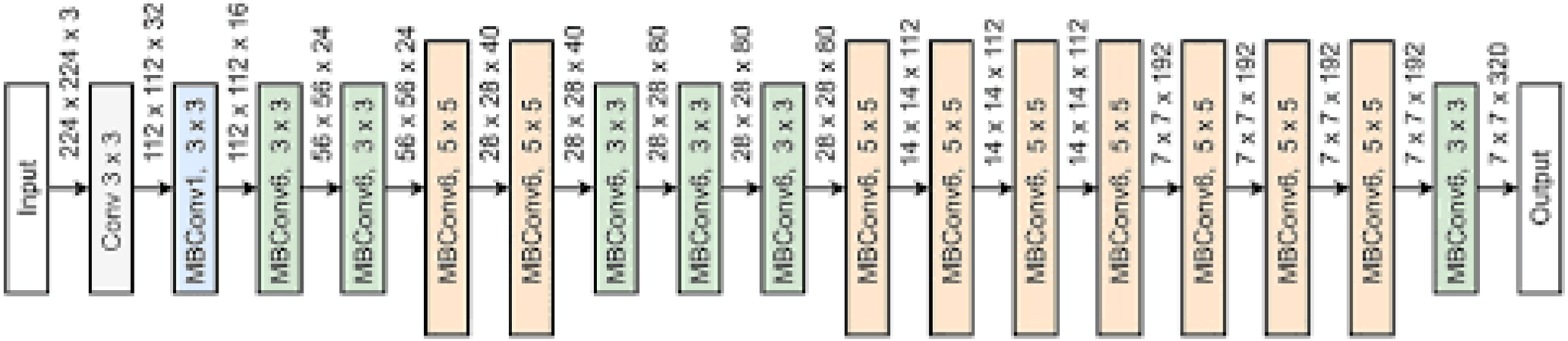

Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0
50
100
150
200
250
300
Epochs

Models Applied

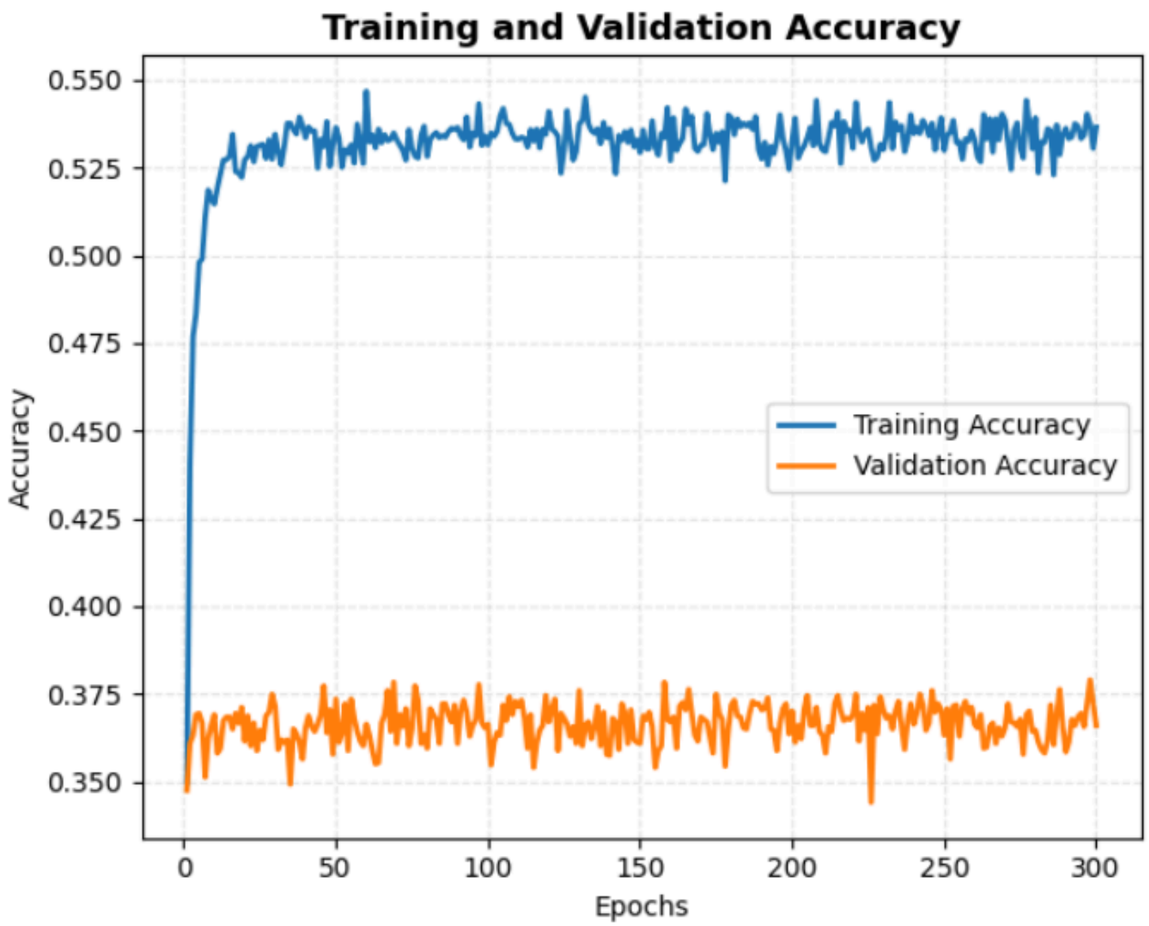
EfficientNetB0

- ~237 layers
- Optimized for high accuracy with fewer parameters

Accuracy → 0.3660
Precision → 0.3420
Recall → 0.3271
F1-Score → 0.3274
AUC → 0.7170



EfficientNetB0

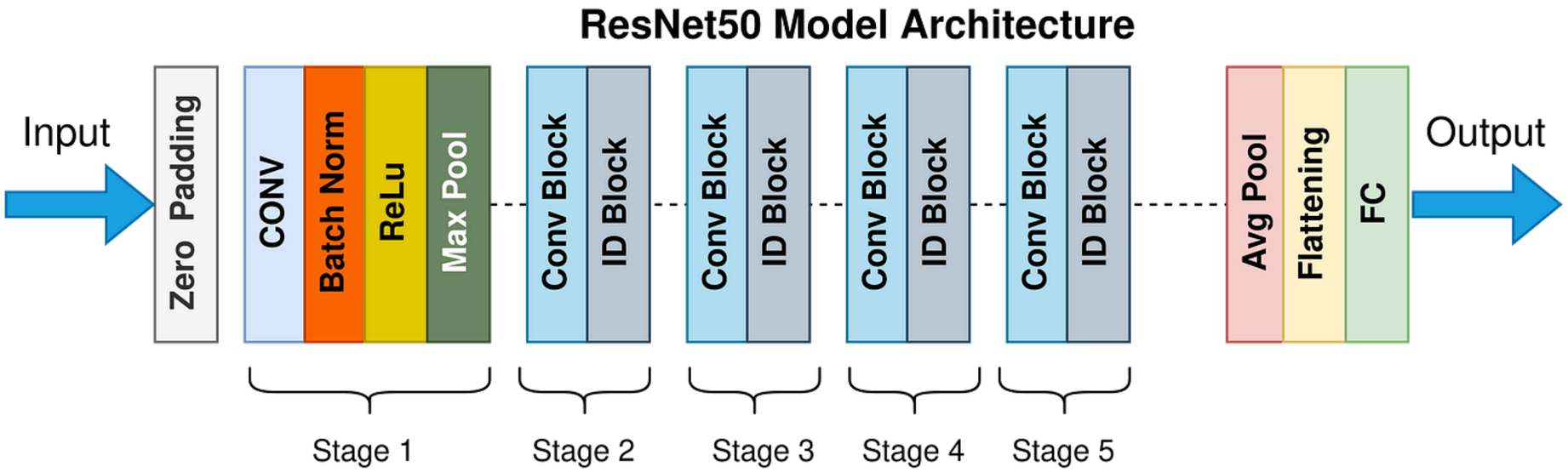


Models Applied

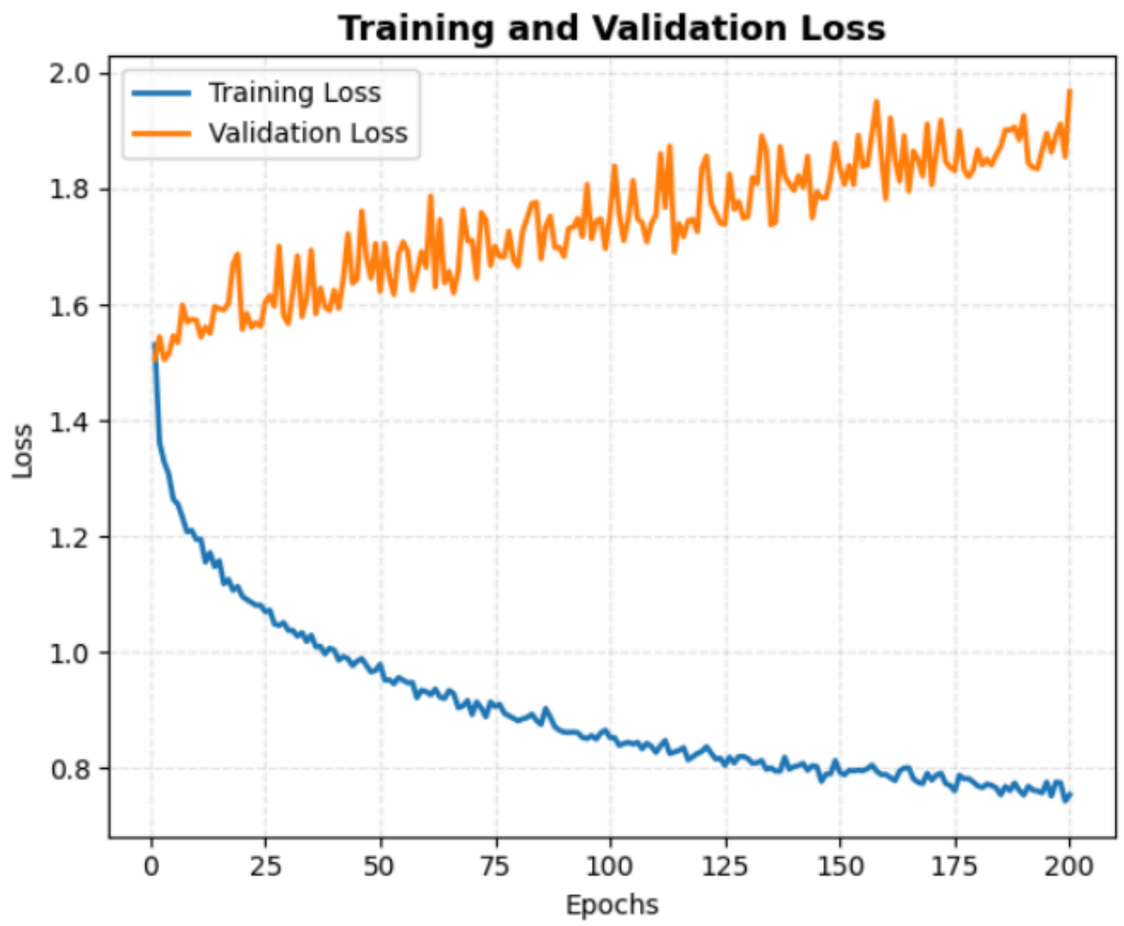
ResNet50

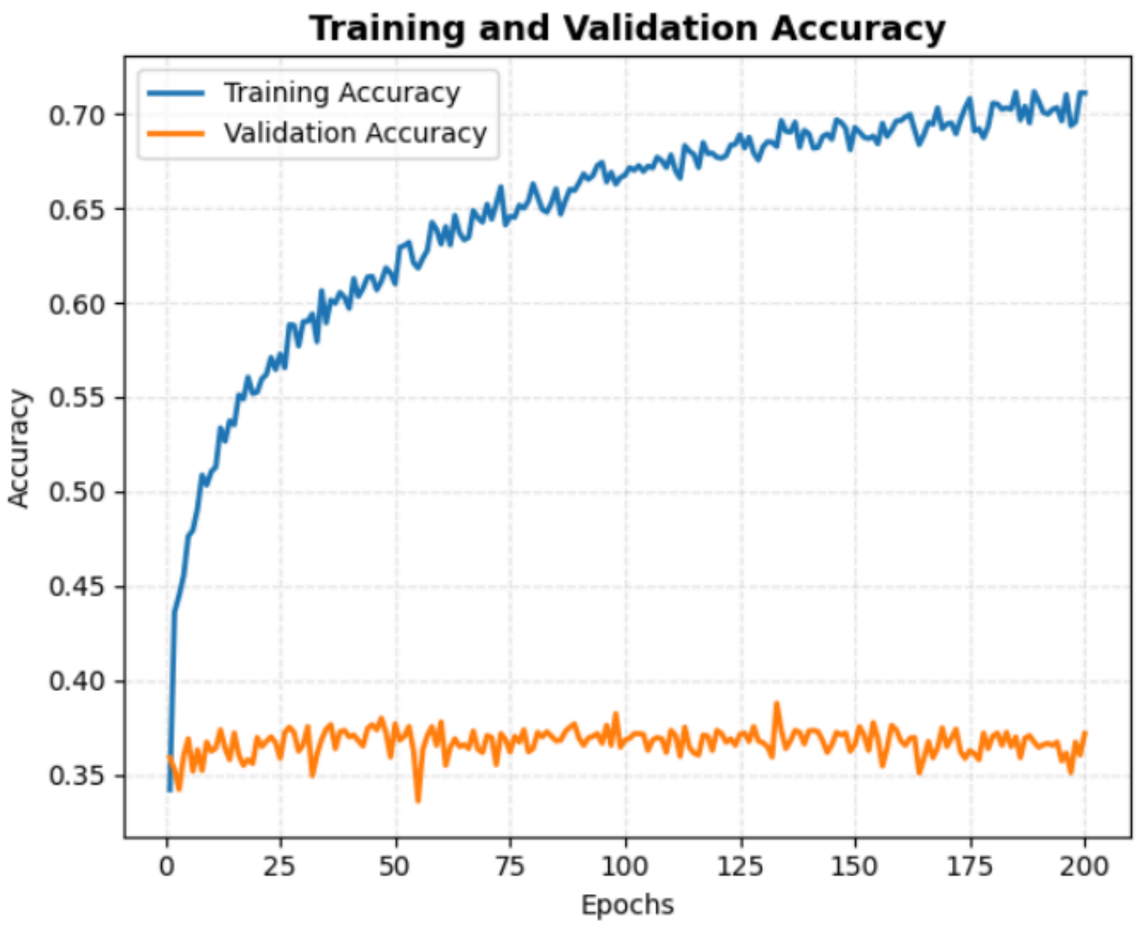
- 50 layers deep
- Uses residual connections to solve vanishing gradient problem

Accuracy → 0.3718
Precision → 0.3890
Recall → 0.3276
F1-Score → 0.3315
AUC → 0.7254



ResNet50

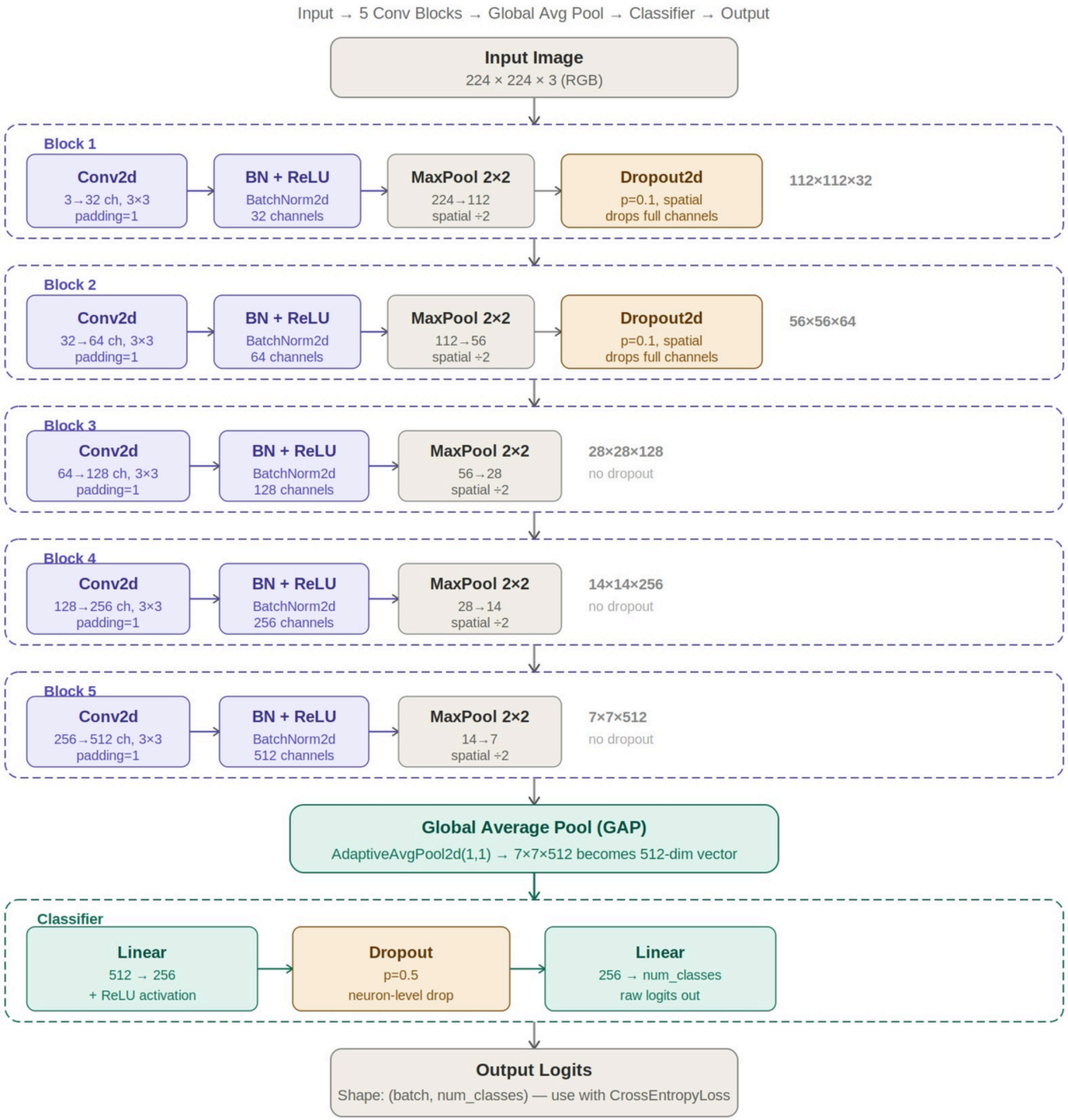

Training and Validation Loss
Training Loss
Validation Loss
Loss
Epochs
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0
25
50
75
100
125
150
175
200


Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0
25
50
75
100
125
150
175
200

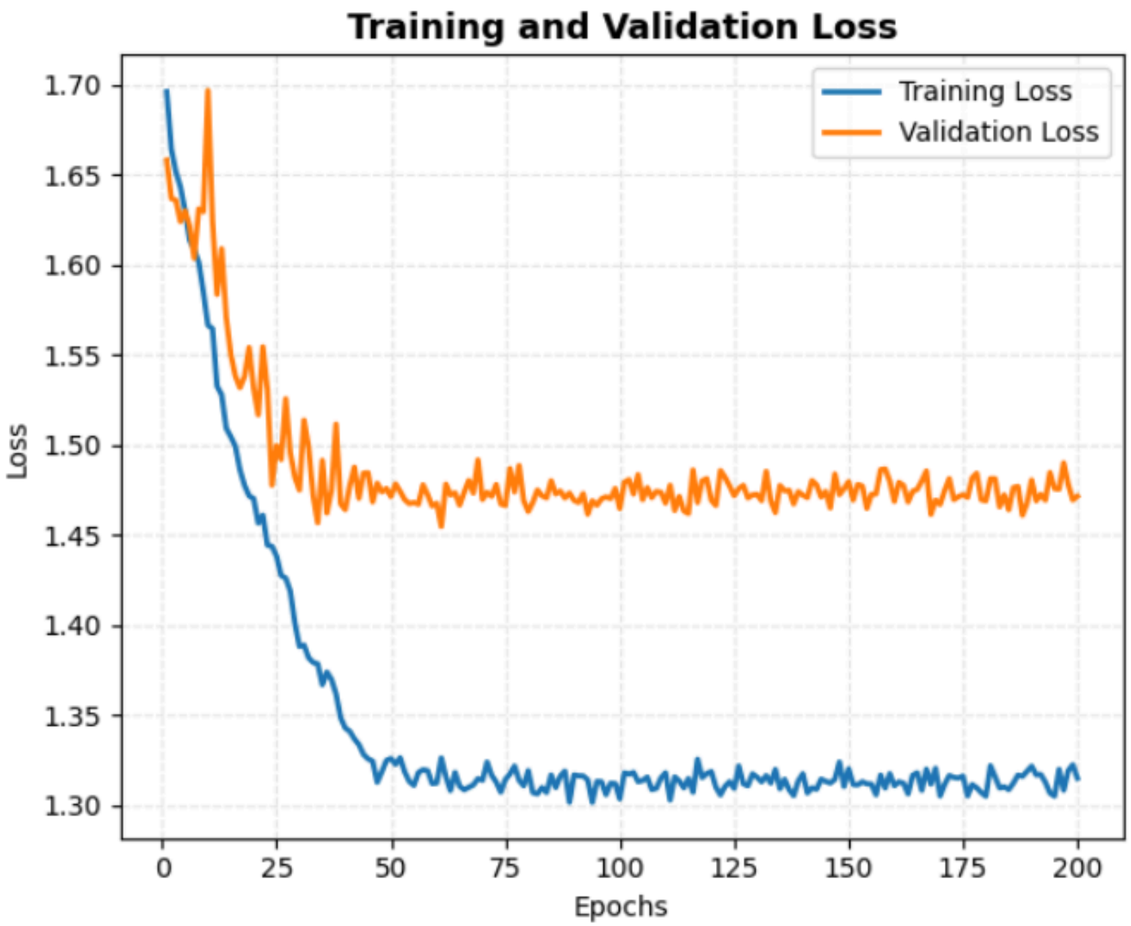
Summary Table

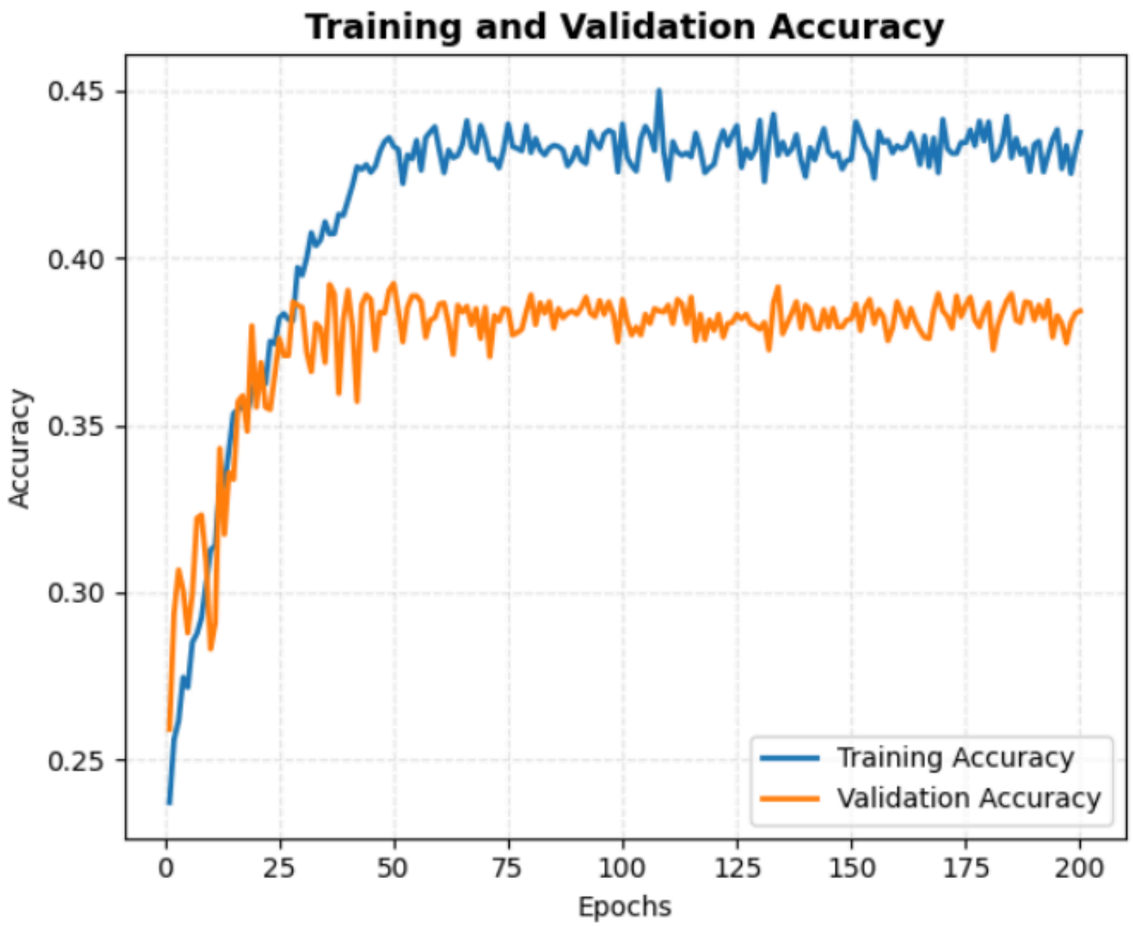
Model	Epochs	Accuracy	Precision	Recall	F1-Score	AUC	Train Time	Test Time
VGG16	200	0.3506	0.31	0.2889	0.2777	0.6831	28078.95s	25.00 s
ResNet50	200	0.3718	0.389	0.3276	0.3315	0.7254	17614.39s	22.20 s
EfficientNetB0	200	0.366	0.342	0.3271	0.3274	0.717	24954.55s	21.73 s
Custom CNN	**200**	**0.3841**	**0.3305**	**0.3084**	**0.2904**	**0.7458**	**18160.94s**	**20.29 s**

CustomCNN


Input → 5 Conv Blocks → Global Avg Pool → Classifier → Output
Input Image
224 × 224 × 3 (RGB)
Block 1
Conv2d
3→32 ch, 3×3
padding=1
BN + ReLU
BatchNorm2d
32 channels
MaxPool 2×2
224→112
spatial ÷2
Dropout2d
p=0.1, spatial
drops full channels
112×112×32
Block 2
Conv2d
32→64 ch, 3×3
padding=1
BN + ReLU
BatchNorm2d
64 channels
MaxPool 2×2
112→56
spatial ÷2
Dropout2d
p=0.1, spatial
drops full channels
56×56×64
Block 3
Conv2d
64→128 ch, 3×3
padding=1
BN + ReLU
BatchNorm2d
128 channels
MaxPool 2×2
56→28
spatial ÷2
28×28×128
no dropout
Block 4
Conv2d
128→256 ch, 3×3
padding=1
BN + ReLU
BatchNorm2d
256 channels
MaxPool 2×2
28→14
spatial ÷2
14×14×256
no dropout
Block 5
Conv2d
256→512 ch, 3×3
padding=1
BN + ReLU
BatchNorm2d
512 channels
MaxPool 2×2
14→7
spatial ÷2
7×7×512
no dropout
Global Average Pool (GAP)
AdaptiveAvgPool2d(1,1) → 7×7×512 becomes 512-dim vector
Classifier
Linear
512 → 256
+ ReLU activation
Dropout
p=0.5
neuron-level drop
Linear
256 → num_classes
raw logits out
Output Logits
Shape: (batch, num_classes) — use with CrossEntropyLoss

CustomCNN


Training and Validation Loss
Training Loss
Validation Loss
Loss
Epochs
1.70
1.65
1.60
1.55
1.50
1.45
1.40
1.35
1.30
0
25
50
75
100
125
150
175
200


Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.45
0.40
0.35
0.30
0.25
0
25
50
75
100
125
150
175
200

Why our model performs better

The custom model is better suited to our PM2.5 dataset because :

1. It fits the data more directly
2. Avoids overfitting from large pretrained backbones
3. Maintains a simpler architecture with strong generalization

Note:
ResNet50 still has higher precision, so the custom CNN is the best overall accuracy winner, while precision behavior may vary by class

Thankyou